Progress Report

- Increment 1 -

Group #2

1) **Team Members**

Colin Maloney – CPM22H – colinpmaloney

Zachary Kupersmith – Zak22 – kupersmithzac

Alex Nino – an23k – alexnino45

Kobus VanSteenburg – kmv21a – kobusthegreat

Samuel Pauly – stp21 – sampauly

2) **Project Title and Description**

Track Tracker. A data platform that detects emerging music tracks before they hit mainstream charts by monitoring Spotify playlists and Soundcloud trending data.

3) **Accomplishments and overall project status during this increment**

a) Established the Database Schema
b) Set up the GitHub Issue Tracker
c) Assigned general roles / tasks
d) Incorporated Spotify & TikTok APIs
e) Set up AWS RDS / PostgreSQL

4) **Challenges, changes in the plan and scope of the project and things that went wrong during this increment**

- We initially planned on using SoundCloud but decided delay the implementation to ensure we reach a minimum viable product as soon as possible.
- We decided TikTok would be a good place to source potentially up and coming songs. We have since incorporated the "Unofficial TikTok API" into our backend. We then plan to cross reference the track titles and fill in the track information with the official Spotify upload.
- Incorporating TikTok required us to rework our Database Schema as well.

5) **Team Member Contribution for this increment**

Colin Maloney contributed to the progress report in all Sections, the implementation, testing and source code regarding the ingestion of Spotify and TikTok API data as well as the data classes which are used to store the API responses. An AWS Account has been set up for the team for the Relational Database Service as well as a configured AWS VPC.

Zachary Kupersmith contributed to the Database schema as well as the Pydantic models for Python backend validation. The PostgreSQL schema has been implemented, and SQL Alchemy is being used for the ORM. . Also contributed to the Spotify API research and section 5 of the Progress report

Kobus Vansteenburg contributed to database structure and creating frontend basic design using Figma which is yet to be implemented. Also contributed to the Spotify API research and section 5 of the Progress report

Samuel Pauly contributed to database structure and TikTok API research. . Also contributed to the Spotify API research and section 5 of the Progress report

Alex Nino contributed to database structure and the implementation of Spotify API Handler source code. . Also contributed to the Spotify API research and section 5 of the Progress report

6) Plans for the next increment

a) Start designing and implementing the front end
b) Ingest TikTok / Spotify Data
c) Enrich the Data
d) Store the Data in AWS

7) Stakeholder Communication

Dear Stakeholders,

We have successfully completed Sprint 1 of the Track-Tracker project.

This sprint focused on foundational architecture and technical validation. The team finalized the initial database schema to support track, artist, and enrichment data. We also researched and tested relevant TikTok and Spotify API endpoints to confirm data availability and integration feasibility.

An AWS account has been set up to host our PostgreSQL database, and the cloud environment configuration is underway. Additionally, we established the base project structure for both the Next.js frontend and the backend services to prepare for feature implementation.

With the core infrastructure and research in place, we are ready to begin building the ingestion and enrichment pipeline in the next sprint.

8) Link to video

https://youtu.be/Ddvcz73BTI4